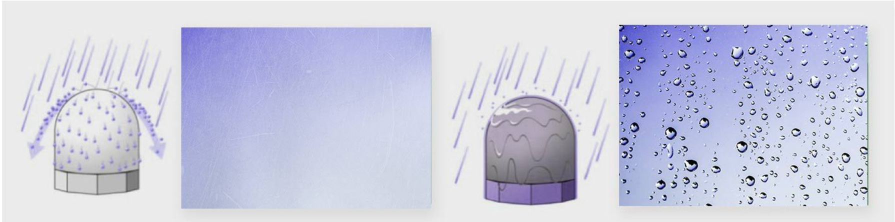
Why Water Repellent



With HIREC   Without HIREC

Prevents water film formation in rainfall reducing radiowave attenuation
Sharply reduces the radio attenuation in antenna, radar, etc., caused by water film formed by rain, sleet or melting snow for a stable transmission quality

"Anti-Stick" characteristic reduces snow and ice accumulation
As well as preventing adhesion of snow and ice, it prevents icicle formation and protects against equipment damage from falling ice and snow.

Self cleaning effect maintains performance for approximately 3 years
The self cleaning effect greatly reduces coating deterioration from air contaminants in outdoor installations. The coating maintains an always fresh surface for about 3 years

NANO COATING REPEAL WATER
AUTO CLEANING DEU TO
TiO2 + UV REACTION

DESIGNED FOR ALL
TELECOMMUNICATION DEVICES

Satellite Internet Solution Provider
From the Heart of Europe.

GLOBALTT

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HIREC PAINT
SUPER HYDROPHOBIC
Water Repellent

GLOBALTT

Protect Your Infrastructural Systems & Assets

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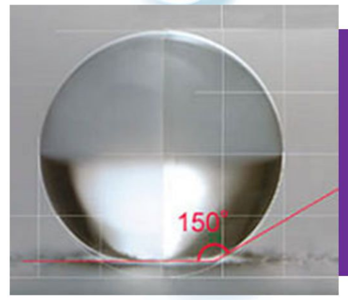
Main Properties & Usage

COUNTER MEASURE

AGAINST WATER

NO MORE WATER ATTENUATION ON TELECOMMUNICATION EQUIPMENT



HIREC is a superhydrophobic water repellent coating material (similar to paint) to protect its critical telecommunications equipment from snow, ice, and rain interference. The coating creates a contact angle of 150 degrees between the surface and the water droplet, virtually eliminating the problems of rain fade and rain, snow, and ice attenuation. This technology has been developed into several products that can be used in a variety of ways, and we are working on creating more.

• With HIREC •

• Rain water runs into drops at the surface, flicking and falling out of the surface.

• No water film is formed.

• Without HIREC •

• Rain water wets the surface and water waves run down the surface.

• Radome surface is covered by a water film.

• MAIN PROPERTIES •

- REPEL THE WATER
- AUTO-CLEANING
- ANTI-WETTING
- ANTI-ICING
- ANTI-CORROSION
- ANTI-MICROBIAL
- AND MANY MORE...

Utilization of HIREC

Without

HIREC

With

HIREC

" HIREC, the hydrophobic coating material, changing the concept of water repellence "

HIREC is highly effective for **preventing water accumulation** on radars, antennas and radomes as a water repellency preventing the formation of water film.

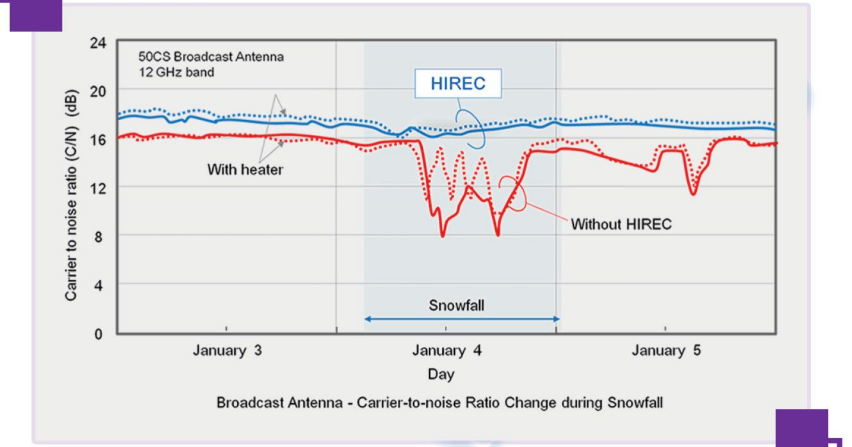
It is also highly effective as a **countermeasure to improve and maintain performance** of antennas, pylons, bridges and towers in snowy or icy climates.

APPLICABLE FOR ALL SECTORS

• ANY TELECOM EQUIPMENT • AVIATION • METEOROLOGICAL RADOMES • MEDICAL • MARITIME • RADAR • ELECTRONIC BOARDS AND COMPONENTS • AND MANY MORE ... •

Changes in Radio Reception of Antennas in Rainfall

Radio receptions of 12 Ghz-Band Broadcast satellite antennas at 5 mm/h rainfall were measured in the field. The aperture diameter of the antenna was 450mm.

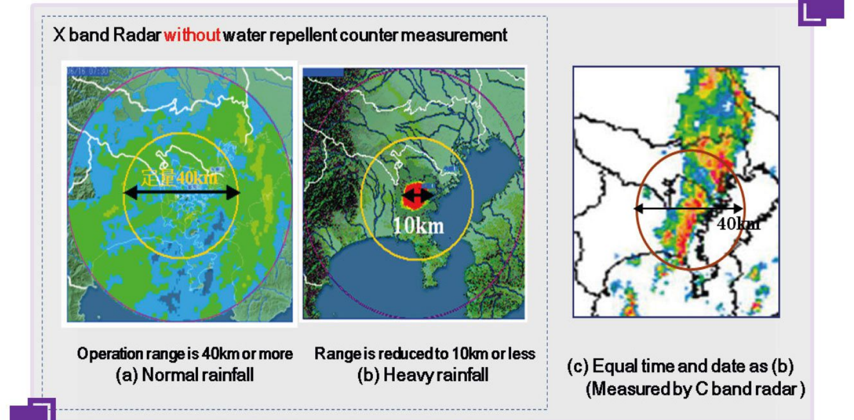


Even a light rain **forming a water film** on the surface of a radome, **can cause signal reduction**.

Notable **signal reduction due to a water film** on the radome during heavy rainfall periods can cause extreme narrowing of the observational range.

For example, a rainfall of 30 mm/hr forming a 0.2 mm thick water film on a radome without HIREC may result in a 5.0 dB attenuation.

Observation range is then narrowed to 10 km compared from 80 km in normal condition.



HIREC's Sterling Photocatalyst Self-Cleaning Mechanism

HIREC integrates photocatalytic material which has strong **UV photooxidation**. The photocatalytic material photooxidizes the oily and organic matters by UV light. The decomposed oily and organic dirt griming grit and dust become easy to peel off and are to be washed out by a rainfall.